Lecture Plan

B.Tech. CSE-Cyber Security

Subject Name: Server Administration 30 hrs +10 experiments

Subject Code:

Sl. No.	Description	Bloom's level*
CO1	Understand server hardware architecture, components, and deployment models used in enterprise environments.	Understand
CO2	Install, configure, and manage server operating systems and virtualization platforms.	Apply
CO3	Implement server roles, system maintenance procedures, and high availability techniques.	Apply
CO4	Configure storage technologies including RAID, backup strategies, and capacity planning.	Analyze
CO5	Apply networking, security controls, disaster recovery, and troubleshooting techniques for server infrastructure.	Analyze

S.No.	Topic	Lecture No.
1.	**Unit 1 – Server Hardware Fundamentals:** **Introduction to Server Hardware** • Enterprise server concepts and deployment models • Server form factors (Tower, Rack, Blade) • Server architecture overview	**Lecture 1**
2.	**Server Components** • CPU architecture and multi-core processors • RAM (ECC/Registered Memory) • Bus types and bus channels • Expansion slots	**Lecture 2**
3.	**Server Storage Components** • Hard drives (HDD, SSD, NVMe) • Network Interface Cards (NICs) • Riser cards • RAID Controllers	**Lecture 3**
4.	**BIOS/UEFI Configuration** • BIOS and UEFI settings • Firmware updates • USB interfaces and ports • Boot configuration	**Lecture 4**

5.	**Power and Cooling Management** • Hot-swappable components • Power requirements and power supply systems • Cooling systems • Airflow management • **Unit Revision**	**Lecture 5**
6.	**Unit 2 – Server Installation and Virtualization: Server Operating System Installation** • Server OS installation planning • Server role identification • Firmware updates during installation	**Lecture 6**
7.	**Server Configuration** • BIOS/UEFI configuration • Disk preparation and partitioning • Hostname configuration • Local account setup	**Lecture 7**
8.	**Network and Domain Configuration** • Network configuration • Domain/Directory joining • Enabling server services • Installing server roles and features	**Lecture 8**
9.	**Applications and Deployment** • Installing applications and drivers • Performance baselining • Unattended installation • Remote installation	**Lecture 9**
10.	**Virtualization Concepts** • Hosts and Guests • Hypervisors • Virtual Machine Management Interfaces • Hardware Compatibility List (HCL) • Resource Allocation	**Lecture 10**
11.	**Unit 3 – Server Roles and Maintenance: Enterprise Server Roles – I** • Web Server • Application Server • Directory Services Server	**Lecture 11**
12.	**Enterprise Server Roles – II** • Database Server • File Server • Print Server • Messaging Server	**Lecture 12**
13.	**Enterprise Server Roles – III** • Mail Server • RRAS Server • Network Services Server • Change Management	**Lecture 13**
14.	**Server Maintenance** • Patch Management • Service Level Agreements (SLA)	**Lecture 14**

	• Outage Management • Performance Monitoring	
15.	**Availability and Documentation** • Hardware Maintenance • Fault Tolerance • High Availability • Asset Management • Infrastructure Documentation • Secure Documentation Storage	**Lecture 15**
16.	**Unit 4 – Storage Technologies:** **Storage Architecture** • DAS, NAS and SAN • Storage Protocols (iSCSI, FCoE) • JBOD	**Lecture 16**
17.	**Storage Devices** • Tape Storage • Optical Drives • Flash Storage • Compact Flash • USB Storage	**Lecture 17**
18.	**Storage Interfaces** • SATA • SAS • NVMe • SCSI • HDD vs SSD	**Lecture 18**
19.	**Storage Management** • Storage deployment • Disk size calculations (Base-10 vs Base-2) • Disk quotas • Compression • Storage capacity planning	**Lecture 19**
20.	**RAID Technologies** • RAID Configuration and RAID Levels • Software vs Hardware RAID • Hot Swap Support • Hot Spare vs Cold Spare • Array Controllers	**Lecture 20**
21.	**Unit 5 – Server Security:** **Physical and System Security** • Physical security methods • Multi-Factor Authentication (MFA) • Server Security Concepts • Server Hardening	**Lecture 21**
22.	**Operating System Security** • OS Hardening • Endpoint Security • Vulnerability Remediation • Hardware Hardening • Environmental Controls	**Lecture 22**

	• Power Protection	
23.	**Network Security Controls** • Firewall • Network Access Control (NAC) • Router ACLs • Network IDS (NIDS) • Authentication Protocols	**Lecture 23**
24.	**Identity and Data Security** • PKI • VPN • IPsec • VLANs • Security Zones • Logical Access Controls • Permissions• ACLs • Storage Encryption • Secure Storage Media Handling & Disposal	**Lecture 24**
25.	**Unit 6 – Networking, Disaster Recovery and Troubleshooting:** **Networking Fundamentals** • IPv4 & IPv6 Addressing • Default Gateway • CIDR Notation • Subnetting • Public & Private Addressing	**Lecture 25**
26.	**Network Services** • Static IP & DHCP • DNS • NetBIOS • WINS • NAT/PAT • MAC Addresses • NIC Configuration	**Lecture 26**
27.	**Network Protocols** • TCP vs UDP • SNMP, SMTP • FTP, FTPS, SFTP, SSH, SCP • HTTP, HTTPS, Telnet • IMAP, POP3, RDP • LDAP, DNS, DHCP, NTP	**Lecture 27**
28.	**Cable Management and Disaster Recovery** • Copper & Fiber Cabling • Connectors and Cable Routing • Disaster Recovery Principles • Site Types • Replication Methods	**Lecture 28**
29.	**Business Continuity and Backup** • Continuity of Operations (COOP) • Backup Methodologies • Backup Media	**Lecture 29**

	• Media Storage Locations	
30.	**Troubleshooting Methodologies** • Hardware Troubleshooting • Software Troubleshooting • Network Troubleshooting • Security Troubleshooting • Storage Troubleshooting	**Lecture 30**

List of Experiments

Experiment No.	Practical Experiment
Experiment 1	Install and configure a server operating system (Windows Server/Linux) in a virtualized environment.
Experiment 2	Configure BIOS/UEFI settings and update server firmware.
Experiment 3	Install and configure server hardware components such as RAM, storage drives, and NICs in a virtual lab or simulator.
Experiment 4	Configure RAID storage (RAID 0, RAID 1, RAID 5, RAID 10) using software RAID tools.
Experiment 5	Configure static IP addressing, DNS, and network connectivity for a server.
Experiment 6	Install and configure server roles and services such as Web Server (Apache/IIS) or File Server.
Experiment 7	Configure and manage virtual machines using Hyper-V, VMware, or VirtualBox.
Experiment 8	Monitor and analyze server performance using system monitoring tools.
Experiment 9	Implement server backup and restore procedures using backup utilities.
Experiment 10	Configure user accounts, groups, and access permissions for secure server administration.

Suggested Readings and References

Sr. No.	Author	Name of the Book	Publisher	Year of Publication
1	Troy McMillan	CompTIA Server+	Willey	2019